Highlights

QuantGhost: Deployment-Robust Weight Steganography and a Capacity–Robustness–Stealth Trilemma

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- Naive mantissa-bit weight-steganography payloads are fully erased by INT8 quantization on real models; a faithful MaleficNet (spread-spectrum) survives INT8, pruning and fine-tuning but only at watermark-scale capacity.
- QuantGhost, a quantization-aware distribution-preserving embedding, reaches one bit per weight surviving INT8, pruning and fine-tuning at a cost of 0.19 percentage points of accuracy on a real ResNet-56.
- It evades the released Model X-Ray detector on a diverse model zoo (0.50, chance) where the prior attack is detected (0.81).
- We prove a capacity-robustness-stealth trilemma: for the neuron-permutation group, any stealthy, quantization-robust, permutation-invariant channel is bounded by $\sum_{\ell} k_h k_w$ bits, a value our construction attains.
- A faithful neuron permutation destroys the high-capacity payload, but a 196-bit invariant key channel survives and bootstraps recovery, so permutation-based sanitization is incomplete.